A Mechanistic Cross-Bridge Model of Cardiac Muscle Mechanics Under Energetic Stress

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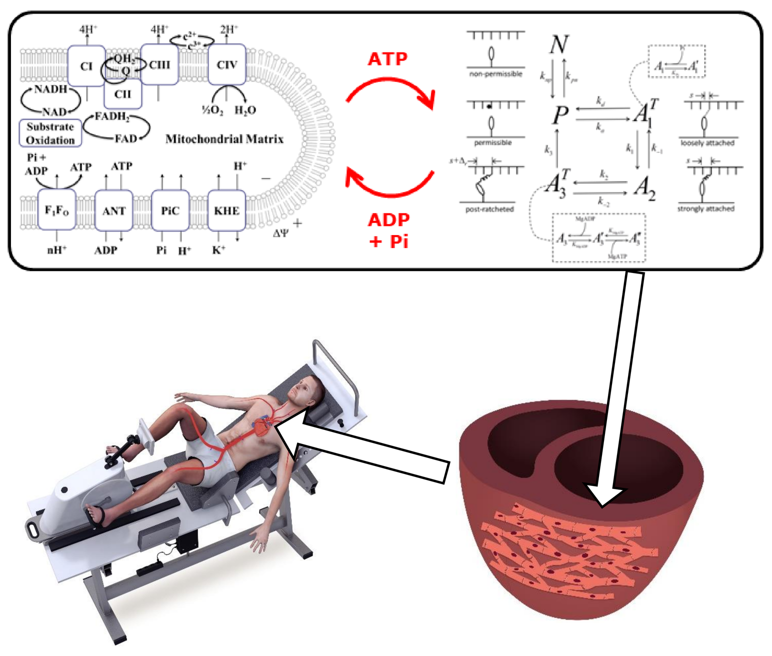
INTRODUCTION

- Heart failure is accompanied by an energetic deficit: impaired mitochondrial function and reduced adenine nucleotide pools lead to lower [ATP] higher [Pi], and reduced PCr/ATP ratio
- These changes directly inhibit the myosin cross-bridge cycling — the molecular engine of cardiac contraction [Collins et al. 2026]
- Beard et al. 2022 established a strain-discretized XB model fit to FV data — but does not reproduce ATP-dependent contractile differences
- Closing this gap requires richer experimental data and additional physical mechanisms in the model



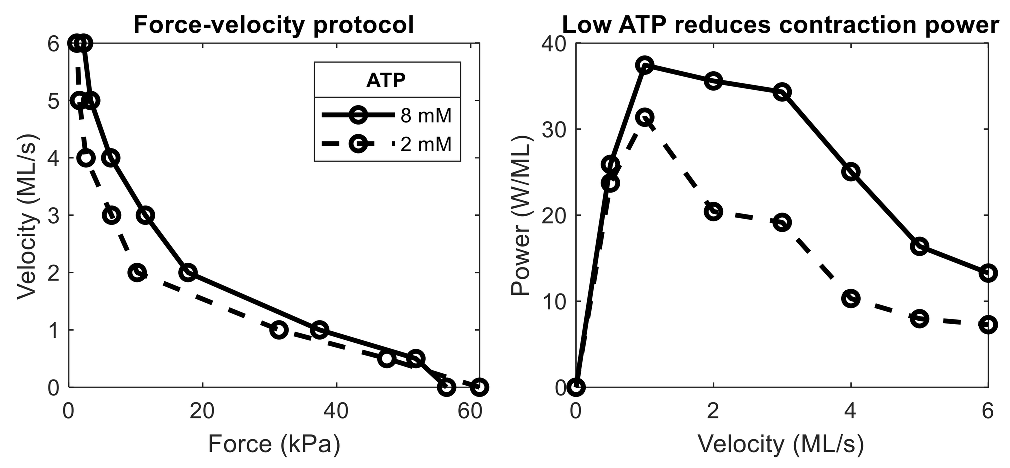
MAIN QUESTIONS

- How does [ATP] modulate sarcomere-level force, velocity, and power stroke kinetics?
- Which specific cross-bridge mechanisms underlie the ATP and Pi sensitivity?
- Can a single mechanistic model simultaneously reproduce force-velocity and slack-restretch behavior across ATP concentrations?
- How does [ATP] affect the muscle efficiency?

ATP MODULATES MUSCLE POWER

Replication of Beard et al. 2022

- Reduced [ATP] slows XB detachment → more attached heads → higher isometric force but slower cycling and reduced peak power output
- FV curve shifts with [ATP]: lower ATP → higher F_0, lower Vmax, reduced peak power
- Beard 2022 model reproduces individual FV curves but does not capture ATP-dependent differences in contractile phenotype
- A nucleotide-explicit state representation is needed to capture [ATP]- and [ADP]-dependent rate changes mechanistically



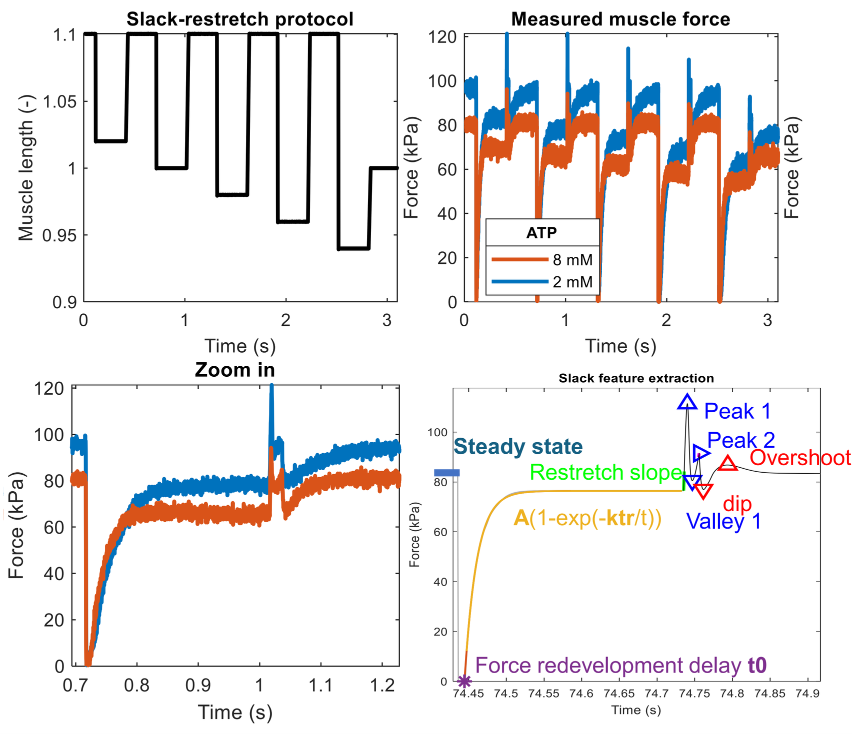
ACQUIRED DATA

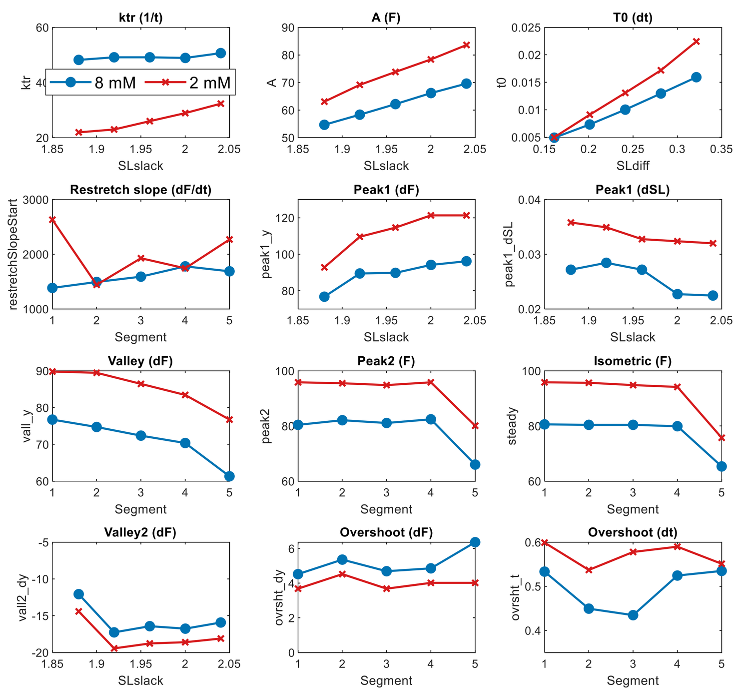
PROTOCOL — Mouse RV Cardiac Trabeculae · Baker Laboratory

- To assess active cross-bridge contributions: rapid release/redevelopment/restretch at SL 2.05 - 1.88 μm and 8 mM and 2 mM ATP.
- To assess passive contributions: block XB cycling in PNB + Mava condition under high Ca^{2+} (right box)

WHAT WE FOUND

- Plateau force ~20 kPa higher at 2 mM than at 8 mM ATP
- ktr depressed at lower [ATP]: slower force redevelopment, slower vmax
- Slack-restretch overshoot and viscoelastic peak systematically altered
- Extracted features: Vmax, F_0, FV shape, ktr, restretch peak, viscoelastic overshoot…

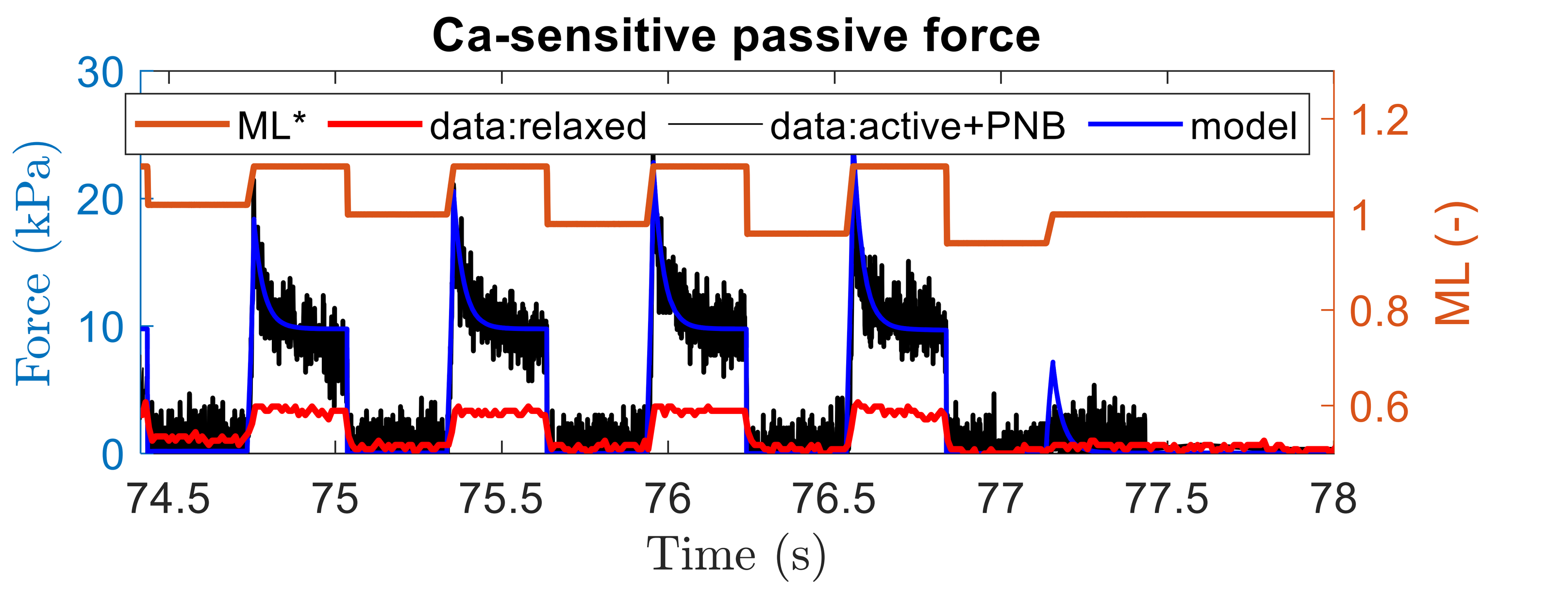




CA-SENSITIVE PASSIVE COMPONENT

Baker, Li, Jezek et al. 2025 (Biophys J) and Jezek, Baker et al. 2025 (J. Physiol.)

- In relaxed condition (no Ca^{2+}) the stress peaks at ~3 kPa following stretch
- In high-Ca^{2+} condition (with PNB+Mava deactivating the XB) the passive force is much higher — a phenomenon attributed to Ca^{2+}-mediated effects on titin [3]
- Naive exponential decay fits the amplitude and time course well enough for this use-case.
- For a detailed model of the passive component, refer to [3]



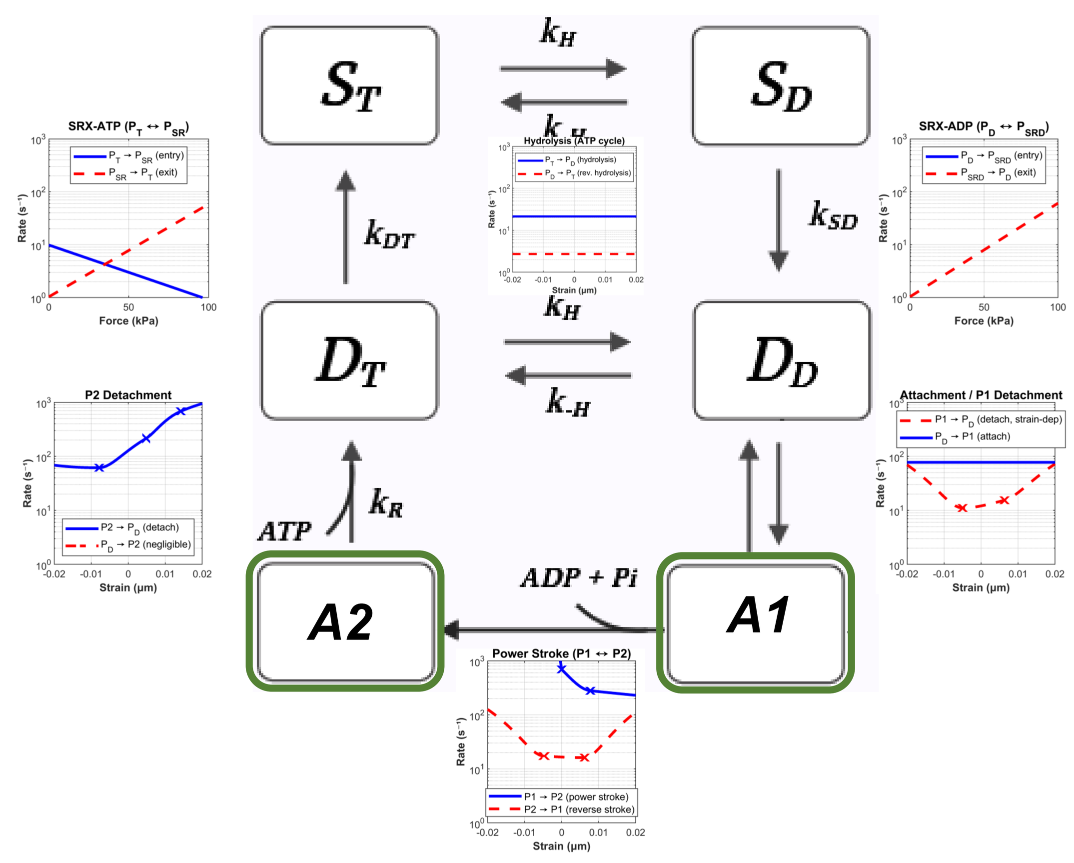
MODEL

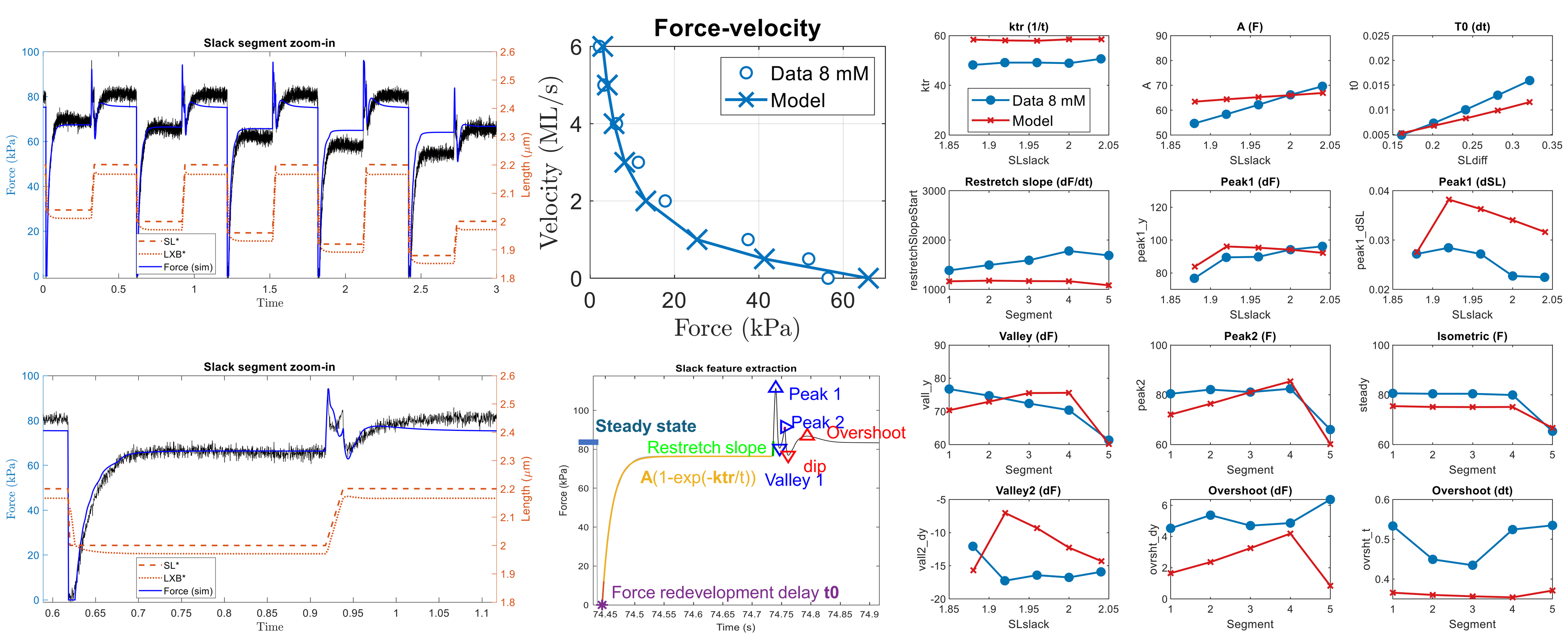
Six-state strain-discretized probabilistic cross-bridge framework

Pool	States	Role
Disordered Relaxed (DRX)	D_T, D_D	Cycling-ready free heads, ATP-bound and ADP·Pi-bound
Attached	A1 (pre), A2 (post-stroke)	Strain-sensitive, force-generating
Super-Relaxed (SRX)	S_T, S_D	Inhibited folded heads, ATP-bound and ADP·Pi-bound

- [ATP], [ADP], and [Pi] modulate transition rates biochemically throughout the full cycle
- Asymmetric piecewise strain-dependent rate modifiers strain dependency of attached-state rate transitions
- Force-dependent SRX mobilization encodes thick filament mechanosensing (Frank-Starling)
- Ca^{2+}-dependent titin viscoelasticity incorporated as parallel passive element [3]

RESULTS — High-ATP (8 mM) model qualitatively reproduces all key features of force-velocity and slack-restretch protocol





CONCLUSION

- Experimental protocol provides a feature-rich training set for model identification
- A six-state XB model qualitatively reproduces key features muscle dynamics observed in training set
- Ca^{2+}-dependent titin stiffening quantified and incorporated as a parallel passive element
- Framework provides a mechanistic bridge between sarcomere mechanics and cardiac energetics

FUTURE WORK

- Quantitative parameter optimization across all ATP and SL conditions
- Explicit coupling to mitochondrial ATP production model
- Sensitivity analysis: which rate steps most impact the energetic phenotype?
- Extension to ADP-dependent contractile modulation and metabolic intervention simulations

References

[1] Collins NL, Dasika S, Van den Bergh F, Bazil J, Beard DA. Systems Analysis of Carboxylate Transport and Oxidation Pathways in Cardiac Mitochondria. bioRxiv 2026.
[2] Beard DA et al. Reduced cardiac muscle power with low ATP simulating heart failure. Biophys. J. 2022.
[3] Jezek F et al. Theoretical analysis of power-law stress relaxation and calcium-dependent passive mechanics in cardiac muscle. J. Physiol. 2026.